

APO-LAMOTRIGINE

Lamotrigine 25 and 100mg Tablets

Antiepileptic

Apo-Lamotrigine 25 mg: each white, shield-shaped, flat-faced, radial edged tablet, scored and engraved "25" on one side, other side "APO" contains 25mg of lamotrigine. Apo-Lamotrigine 100 mg: each peach, shield-shaped, flat-faced, bevelled edged tablet, scored and engraved "LAM over 100" on one side, other side "APO" contains 100mg of lamotrigine.

Clinical Pharmacology

Mode of action

The results of pharmacological studies suggest that lamotrigine is a use-dependent blocker of voltage gated sodium channels. It produces a use- and voltage dependent block of sustained repetitive firing in cultured neurons and inhibits pathological release of glutamate (the amino acid which plays a key role in the generation of epileptic seizures), as well as inhibiting glutamate-evoked bursts of action potentials.

Pharmacodynamics

In tests designed to evaluate the central nervous system effects of drugs, the results obtained using doses of 240 mg lamotrigine administered to healthy volunteers did not differ from placebo, whereas both 1000 mg phenytoin and 10 mg diazepam each significantly impaired fine visual motor co-ordination and eye movements, increased body sway and produced subjective sedative effects.

In another study, single oral doses of 600mg carbamazepine significantly impaired fine visual motor co-ordination and eye movements, while increasing both body sway and heart rate, whereas results with lamotrigine at doses of 150mg and 300mg did not differ from placebo.

Pharmacokinetics properties

Lamotrigine is rapidly and completely absorbed from the gut with no significant first pass metabolism. Peak plasma concentrations occur approximately 2.5 hours after oral drug administration. Time to maximum concentration is slightly delayed after food but the extent of absorption is unaffected. The pharmacokinetics are linear up to 450mg, the highest single dose tested. There is considerable inter-individual variation in steady state maximum concentrations but within an individual concentrations vary very little.

Binding to plasma proteins is about 55%. It is very unlikely that displacement from plasma proteins would result in toxicity. The volume of distribution is 0.92 to 1.22 L/kg.

The mean steady state clearance in healthy adults is 39 ± 14 mL/min. Clearance of lamotrigine is primarily metabolic with subsequent elimination of glucuronide-conjugated material in urine. Less than 10% is excreted unchanged in the urine. Only about 2% of drug-related material is excreted in faeces. Clearance and half-life are independent of dose. The mean elimination half-life in healthy adults is 24 to 35 hours. UDP-glucuronyl transferases have been identified as the enzymes responsible for metabolism of lamotrigine. In a study of subjects with Gilbert's Syndrome, mean apparent clearance was reduced by 32% compared with normal controls but the values are within the range for the general population.

Lamotrigine induces its own metabolism to a modest extent depending on dose. However, there is no evidence that lamotrigine affects the pharmacokinetics of other antiepileptic drug (AED) and data suggest that interactions between lamotrigine and drugs metabolised by cytochrome P450 enzymes are unlikely to occur.

The half-life of lamotrigine is greatly affected by concomitant medication. Mean half-life is reduced to approximately 14 hours when given with enzyme-inducing drugs such as carbamazepine and phenytoin and is increased to a mean of approximately 70 hours when co-administered with sodium valproate alone. (see Posology and Method of Administration).

Clearance adjusted for bodyweight is higher in children aged 12 years and under than in adults with the highest values in children under five years. The half-life of lamotrigine is generally shorter in children than in adults with a mean value of approximately 7 hours when given with enzyme-inducing drugs such as carbamazepine and phenytoin and increasing to mean values of 45 to 50 hours when co-administered with sodium valproate alone (see Posology and Method of Administration).

INDICATIONS AND CLINICAL USE

Apo-Lamotrigine is indicated for use as adjunctive or monotherapy in the treatment of epilepsy, for partial seizures and generalized seizures, including tonic-clonic seizures, and the seizures associated with Lennox-Gastaut Syndrome.

DOSAGE AND ADMINISTRATION

Dosage in monotherapy

Adults and children over 12 years (see Table 1) The initial Apo-Lamotrigine dose in monotherapy is 25mg once a day for two weeks, followed by 50mg once a day for two weeks. Thereafter, the dose should be increased by a maximum of 50mg-100mg every 1-2 weeks until the optimal response is achieved. The usual maintenance dose to achieve optimal response is 100 - 200mg/day given once a day or as two divided doses. Some patients have required 500mg/day of Lamotrigine to achieve the desired response.

The initial dose and subsequent dose escalation should not be exceeded to minimise the risk of rash (see Special Warnings and Special Precautions for Use).

Dosage in add-on therapy

Adults and children over 12 years (see Table 1)

In patients taking valproate with / without any other AED, the initial Lamotrigine dose is 25 mg every alternate day for two weeks, followed by 25 mg once a day for two weeks. Thereafter, the dose should be increased by a maximum of 25-50mg every 1-2 weeks until the optimal response is achieved. The usual maintenance dose to achieve optimal response is 100- 200mg/day given once a day or in two divided doses.

In those patients taking enzyme inducing AEDs with/ without other AEDs (except valproate), the initial Lamotrigine dose is 50 mg once a day for two weeks, followed by 100 mg/day given in two divided doses for two weeks. Thereafter, the dose should be increased by a maximum of 100mg every 1-2 weeks until the optimal response is achieved. The usual maintenance dose to achieve optimal response is 200-400mg/day given in two divided doses. Some patients have required 700 mg/day of Lamotrigine to achieve the desired response.

In patients taking AEDs where the pharmacokinetic interaction with lamotrigine is currently not known, the dose escalation as recommended for lamotrigine with concurrent valproate should be used, thereafter, the dose should be increased until optimal response is achieved.

Table 1 Recommended treatment regimen for adults and children over 12 years of age

Treatment regimen		Weeks 1 + 2	Weeks 3 + 4	Usual Maintenance Dose
Monotherapy		25 mg (once a day)	50 mg (once a day)	100 – 200 mg (once a day or two divided doses) To achieve maintenance, doses may be increased by 50 – 100 mg every one to two weeks
Add-on therapy with valproate regardless of any concomitant medications		12.5 mg (given 25 mg on alternate days)	25 mg (once a day)	100 – 200 mg (once a day or two divided doses) To achieve maintenance, doses may be increased by 25 – 50 mg every one to two weeks
Add-on therapy without valproate	This dosage regimen should be used with: phenytoin carbamazepine phenobarbital primidone or with other inducers of lamotrigine glucuronidation (see Drug Interactions).	50 mg (once a day)	100 mg (two divided doses)	200 – 400 mg (two divided doses) To achieve maintenance, doses may be increased by 100 mg every one to two weeks

Note: In patients taking AEDs where the pharmacokinetic interaction with lamotrigine is currently not known (see section Drug Interactions), the treatment regimen as recommended for lamotrigine with concurrent valproate should be used, thereafter, the dose should be increased until optimal response is achieved.

The initial dose and subsequent dose escalation should not be exceeded to minimise the risk of rash (see Special Warnings and Precautions).

Women and Hormonal Contraceptives (see Special Warnings and Precautions)

(a) Starting lamotrigine in patients taking hormonal contraceptives

Dose escalation should follow the guidelines recommended in Table 1 above.

(b) Starting hormonal contraceptives in patients taking lamotrigine

For women NOT taking inducers of lamotrigine glucuronidation such as phenytoin, carbamazepine, phenobarbital, primidone or rifampicin, the maintenance dose of lamotrigine may need to be increased by as much as two-fold, according to clinical response. For women taking lamotrigine in addition to inducers of lamotrigine glucuronidation, adjustment may not be necessary.

(c) Stopping hormonal contraceptives in patients taking lamotrigine

For women NOT taking inducers of lamotrigine glucuronidation the maintenance dose of lamotrigine may need to be decreased by as much as 50%, according to clinical response.

For women taking lamotrigine in addition to inducers of lamotrigine glucuronidation, adjustment may not be necessary.

Pregnancy and *post-partum*

Dose adjustment may be necessary during pregnancy and post-partum.

Elderly

No dosage adjustment from recommended schedule is required. The pharmacokinetics of lamotrigine in this age group do not differ significantly from a non-elderly population.

Hepatic Impairment

Initial, escalation and maintenance doses should generally be reduced by approximately 50% in patients with moderate (Child-Pugh grade B) and 75% in severe (Child-Pugh grade C) hepatic impairment. Escalation and maintenance doses should be adjusted according to clinical response.

MODE OF ADMINISTRATION

Apo-Lamotrigine Tablets should be swallowed whole with a little water.

To ensure a therapeutic dose is maintained the weight of a child must be monitored and the dose reviewed as weight changes occur.

If a calculated dose of lamotrigine (e.g. for use in children and patients with hepatic impairment) does not equate to whole tablets the dose to be administered is that equal to the lower number of whole tablets.

When concomitant antiepileptic drugs are withdrawn to achieve Apo-Lamotrigine monotherapy or other antiepileptic drugs (AEDs) are added-on to treatment regimes containing Apo-Lamotrigine consideration should be given to the restarted in patients with previous hypersensitivity (see section 4.3).

Rash has also been reported as part of a hypersensitivity syndrome associated with a effect this may have on lamotrigine pharmacokinetics (see 4.5 Interaction with other Medicinal Products and other Forms of Interaction).

Restarting Therapy

Prescribers should assess the need for escalation to maintenance dose when restarting lamotrigine in patients who have discontinued lamotrigine for any reason, since the risk of serious rash is associated with high initial doses and exceeding the recommended dose escalation for lamotrigine (see section 4.4). The greater the interval of time since the previous dose, the more consideration should be given to escalation to the maintenance dose. When the interval since discontinuing lamotrigine exceeds five half-lives (see section 5.2), lamotrigine should generally be escalated to the maintenance dose according to the appropriate schedule, as though initiating therapy (see section 4.2).

CONTRAINDICATIONS

Apo-Lamotrigine is contraindicated in individuals with known hypersensitivity to lamotrigine.

SPECIAL WARNINGS AND PRECAUTIONS

Serious warnings & precautions

Serious skin rashes

- Serious and sometimes fatal skin rashes, including Stevens-Johnson Syndrome (SJS) and Toxic Epidermal Necrolysis (TEN), have occurred with the use of lamotrigine. The incidence of these rashes in clinical trials was 1% (1/100) in pediatric patients (age < 16 years) and 0.3% (3/1000) in adults.
- In children, the initial presentation of a rash can be mistaken for an infection. Physicians should consider the possibility of a drug reaction in children that develop symptoms of rash and fever during the first eight weeks of therapy.
- Risk of rapid titration: Do not exceed the recommended initial dose and subsequent dose escalations of Apo-Lamotrigine. A higher incidence of serious skin rashes has been associated with more rapid initial titration (exceeding the recommended initial dose or exceeding the recommended dose escalation).
- Risk of concomitant use with valproic acid.
- Risk with history of rash to other antiepileptics.
- Nearly all cases of rash associated with lamotrigine have occurred within 2 to 8 weeks of treatment initiation. However, isolated cases have been reported after prolonged treatment (e.g., 6 months). Accordingly, duration of therapy cannot be relied upon as a means to predict the potential risk signalled by the first appearance of a rash.
- Although benign rashes also occur with lamotrigine, it is not possible to predict reliably which rashes will prove to be life threatening. Accordingly, all patients who develop rash should be promptly evaluated and Apo-Lamotrigine withdrawn immediately, unless the rash is clearly not drug related.

Acute Multi-Organ Failure: Multi-organ failure, which in some cases has been fatal or irreversible, has been observed in patients receiving lamotrigine. Fatalities associated with multi-organ failure and various degrees of hepatic failure have been reported in adult patients and pediatric patients who received lamotrigine in epilepsy clinical trials. No such fatalities have been reported in bipolar patients in clinical trials. Rare fatalities from multi-organ failure have also been reported in compassionate plea and postmarketing use. The majority of these deaths occurred in association with other serious medical events, including status epilepticus and overwhelming sepsis, and hantavirus, making it difficult to identify the initial cause.

Additionally, few patients developed multi-organ dysfunction and disseminated intravascular coagulation 9 to 14 days after lamotrigine was added to their AED regimens. Rash and elevated transaminases were also present in all patients and rhabdomyolysis was noted in few patients. Both pediatric patients were receiving concomitant therapy with valproate, while the adult patient was being treated with carbamazepine and clonazepam. All patients subsequently recovered with supportive care after treatment with lamotrigine was discontinued.

Dihydrofolate Reductase: Lamotrigine was shown to be a weak inhibitor of dihydrofolate reductase *in vitro*, hence there is a possibility of interference with folate metabolism during long-term therapy. In clinical studies, lamotrigine did not affect blood folate concentrations or associated hematologic parameters.

Drug Discontinuation: Abrupt discontinuation of any AED in a responsive patient with epilepsy may provoke rebound seizures. In general, withdrawal of an AED should be gradual to minimize this risk. Unless safety concerns (i.e. rash) require a more rapid withdrawal, the dose of lamotrigine should be tapered over a period of at least two weeks.

Estrogen-Containing Product Including Hormonal Contraceptives: Patients taking Apo-Lamotrigine should be advised not to start or stop their estrogen-containing products including oral contraceptives and hormonal replacement therapy (HRT) without consulting their physician. Close clinical monitoring and significant adjustments in the maintenance dose of lamotrigine may be required. Gradual transient increases in lamotrigine levels may occur during the week of no active hormone preparation (pill-free week), as much as 2-fold at the end of the week, for women not also taking a drug that increases the clearance of lamotrigine

Interaction with Antiepileptic Drugs: In a multidrug regimen, antiepileptic drugs may affect lamotrigine clearance, and dosage adjustment of Apo-Lamotrigine might be required.

Other Products containing Lamotrigine: Lamotrigine tablets should not be administered to patients currently being treated with any other preparation containing lamotrigine without consulting a doctor.

Potential Medication Errors: Medication errors involving lamotrigine have occurred. In particular, the name lamotrigine can be confused with the names of other commonly used medications. To reduce the potential of medication errors, write and say Apo-Lamotrigine clearly. To avoid the medication error of using the wrong drug, patients should be strongly advised to visually inspect their tablets to verify that they are Apo-Lamotrigine, each time they fill their prescription.

Patients with Special Diseases and Conditions: Clinical experience with lamotrigine in patients with concomitant illness is limited. Caution is advised when using Apo-Lamotrigine in patients with diseases or conditions that could affect the metabolism or elimination of the drug.

Brugada-type ECG: A very rare association with Brugada-type ECG has been observed, although a causal relationship has not been established. Therefore careful consideration should be given before using Apo-Lamotrigine in patients with Brugada syndrome.

Cardiac Rhythm Conduction Abnormalities: Apo-Lamotrigine should be used with caution in patients with cardiac conduction abnormalities, and in patients taking concomitant medications which depress AV conduction. Lamotrigine could potentially slow ventricular conduction (by causing QRS prolongation) and induce proarrhythmia. This can lead to sudden death in patients with clinically significant structural or functional heart disease such as heart failure, valvular heart disease, congenital heart disease, conduction system disease, ventricular arrhythmias, cardiac channelopathies (e.g., Brugada syndrome), clinically important ischemic heart disease, or multiple risk factors for coronary artery disease. Concomitant use of other sodium channel blockers (e.g., some other anti-epileptic medications) may further increase the risk of proarrhythmia. Therefore, any expected or observed benefit of lamotrigine for those patients must be carefully weighed against the potential risks for serious or fatal cardiac events.

Driving and Operating Machinery: Patients with uncontrolled epilepsy should not drive or handle potentially dangerous machinery. During clinical trials common adverse effects included dizziness, ataxia, drowsiness, diplopia, and blurred vision. Patients should be advised to refrain from activities requiring mental alertness or physical coordination until they are sure that Apo-Lamotrigine does not affect them adversely.

Blood Dyscrasias: There have been reports of blood dyscrasias that may or may not be associated with the hypersensitivity syndrome. These have included neutropenia, leukopenia, anemia, thrombocytopenia, pancytopenia, and, rarely, aplastic anemia and pure red cell aplasia.

Hepatic/Biliary/Pancreatic: A reduced dosage should be used for all hepatically impaired patients, and Apo-Lamotrigine should be administered with caution particularly in those patients with severe hepatic impairment.

Immune: Hemophagocytic lymphohistiocytosis (HLH) has occurred in patients taking lamotrigine. HLH is a syndrome of pathological immune activation which can be life threatening, characterized by clinical signs and symptoms such as fever, rash, neurological symptoms, hepatosplenomegaly, lymphadenopathy, cytopenias, high serum ferritin, hypertriglyceridaemia and abnormalities of liver function and coagulation. Symptoms occur generally within 4 weeks of treatment initiation. Immediately evaluate patients who develop these signs and symptoms and consider a diagnosis of HLH. Lamotrigine should be discontinued unless an alternative aetiology can be established.

Hypersensitivity: The 100 mg tablet coating contains an azo dye (FD&C Yellow No. 6 aluminium lake) which may cause allergic reactions.

Monitoring and Laboratory Tests: The relationship between clinical efficacy and plasma concentrations has not been clearly established. Based on the possible pharmacokinetic interactions between lamotrigine and other drugs including AEDs, monitoring of the plasma levels of lamotrigine and concomitant drugs may be indicated, particularly during dosage adjustments. In general, clinical judgment should be exercised regarding monitoring of plasma levels of lamotrigine and other drugs and whether or not dosage adjustments are necessary.

Aseptic Meningitis: Therapy with lamotrigine increases the risk of developing aseptic meningitis. Because of the potential for serious outcomes of untreated meningitis due to other causes, patients should also be evaluated for other causes of meningitis and treated as appropriate.

Postmarketing cases of aseptic meningitis have been reported in pediatric and adult patients taking lamotrigine for various indications. Symptoms upon presentation have included headache, fever, nausea, vomiting, and nuchal rigidity.

Rash photophobia, myalgia, chills, altered consciousness, and somnolence were also noted in some cases. Symptoms have been reported to occur within 1 to 40 days following the initiation of treatment. In most cases, symptoms were reported to resolve after discontinuation of lamotrigine. Re-exposure resulted in a rapid return of symptoms (from within 30 minutes to 1 day following re-initiation of treatment) that were frequently more severe. Some of the patients treated with lamotrigine who developed aseptic meningitis had underlying diagnoses of systemic lupus erythematosus or other autoimmune diseases. Lamotrigine should not be restarted in patients who have discontinued due to aseptic meningitis associated with prior treatment of lamotrigine.

Cerebrospinal fluid (CSF) analysed at the time of clinical presentation in reported cases was characterized by a mild to moderate pleocytosis, normal glucose levels, and mild to moderate increase in protein. CSF white blood cell count differentials showed a predominance of neutrophils in a majority of the cases, although a predominance of lymphocytes was reported in approximately one third of the cases. Some patients also had new onset of signs and symptoms of involvement of other organs (predominantly hepatic and renal involvement) which may suggest that in these cases the aseptic meningitis observed was part of a hypersensitivity reaction.

Status Epilepticus: There are reports in the literature that severe convulsive seizures including status epilepticus may lead to rhabdomyolysis, multi-organ dysfunction and disseminated intravascular coagulation, sometimes with fatal outcome. Similar cases have occurred in association with the use of lamotrigine.

Suicidal Behaviour and Ideation: Symptoms of depression and /or bipolar disorder may occur in patients with epilepsy, and there is evidence that patients with epilepsy and bipolar disorder have an elevated risk for suicidality.

25 to 50% of patients with bipolar disorder attempt suicide at least once and may experience worsening of their depressive symptoms and/or the emergence of suicidal ideation and behaviours (suicidality) whether or not they are taking medications for bipolar disorder, including lamotrigine.

Suicidal ideation and behaviour have been reported in patients treated with antiepileptic agents in several indications. All patients treated with antiepileptic drugs, irrespective of indication, should be monitored for signs of suicidal ideation and behaviour and appropriate treatment should be considered. Patients (and caregivers of patients) should be advised to seek medical advice should signs of suicidal ideation or behaviour emerge.

An FDA meta-analysis of randomized placebo controlled trials, in which antiepileptic drugs were used for various indications, has shown a small increased risk of suicidal ideation and behaviour in patients treated with these drugs. The mechanism of this risk is not known, and the available data do not exclude the possibility of an increased risk for lamotrigine.

Renal: A study in individuals with chronic renal failure (not receiving other AEDs) indicated that the elimination half-life of unchanged lamotrigine is prolonged relative to individuals with normal renal function. Apo-Lamotrigine should be used with caution in patients with severe renal impairment.

Skin: Serious and sometimes fatal skin rashes, including Stevens-Johnson Syndrome (SJS) and Toxic Epidermal Necrolysis (TEN), have occurred with the use of lamotrigine.

Drug Reaction with Eosinophilia and Systemic Symptoms: Multi-organ hypersensitivity reactions, also known as Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS), have occurred with lamotrigine. Some have been fatal or life threatening. DRESS typically, although not exclusively, presents with fever, rash, facial oedema, and/or lymphadenopathy in association with other organ system involvement, such as hepatitis, nephritis, hematologic abnormalities, aseptic meningitis, myocarditis, or myositis, sometimes resembling an acute viral infection. Eosinophilia is often present. This disorder is variable in its expression and other organ systems not noted here may be involved. The syndrome shows a wide spectrum of clinical severity and may rarely lead to disseminated intravascular coagulation (DIC) and multi-organ failure.

It is important to note that early manifestations of hypersensitivity (e.g., fever, lymphadenopathy) may be present even though a rash is not evident. If such signs or symptoms are present, the patient should be evaluated immediately. Apo-Lamotrigine should be discontinued if an alternative etiology for the signs or symptoms cannot be established.

Prior to initiation of treatment with Apo-Lamotrigine, the patient should be instructed that a rash or other signs or symptoms of hypersensitivity (e.g., fever, lymphadenopathy) may herald a serious medical event and that the patient should report any such occurrence to a physician immediately.

Asian Ancestry and Allelic variation in HLA-B: Some meta-analyses have shown that human leukocyte antigen (HLA)-B*1502 allele in individuals of Asian (primarily Han Chinese and Thai†) origin is associated with the risk of developing SJS/TEN when treated with lamotrigine. If patients are known to be carrying this genetic variant, the

benefits of treatment with lamotrigine should be carefully weighed against the risk of developing SJS/TEN. Should signs and symptoms suggest a severe skin reaction such as SJS or TEN, lamotrigine should be withdrawn at once, under clinical supervision.

Many HLA-B*1502-positive Asian patients treated with lamotrigine will not develop SJS/TEN, and these reactions can still occur infrequently in HLA-B*1502-negative patients of any ethnicity. The role of other possible factors in the development of, and morbidity from, SJS/TEN, such as AED dose, compliance, concomitant medications, co-morbidities, and the level of dermatologic monitoring have not been studied.

In addition, it should be kept in mind that the majority of lamotrigine -treated patients who will experience SJS/TEN have this reaction within the first few months of treatment.

Photosensitivity: There have also been reports of photosensitivity reactions associated with lamotrigine use. If lamotrigine-associated photosensitivity is suspected in a patient showing signs of photosensitivity (such as an exaggerated sunburn), treatment discontinuation should be considered. If continued treatment with lamotrigine is considered clinically justified, the patient should be advised to avoid exposure to sunlight and artificial UV light and take protective measures (e.g. use of protective clothing and sunscreens).

Skin-related events: Serious rashes requiring hospitalisation, including rare reports of fatalities, have occurred with lamotrigine. The risk of serious skin rashes is higher in children than in adults. In adult controlled studies of adjunctive lamotrigine therapy, the incidence of rash (usually maculopapular and/or erythematous) in patients receiving lamotrigine was higher compared with placebo patients. The rash usually occurred within the first eight weeks of therapy and resolved during continued administration of lamotrigine. The rate of rash-related withdrawal in clinical studies was higher with more rapid initial titration dosing, and in patients receiving concomitant valproic acid (VPA), particularly in the absence of AEDs that induce lamotrigine glucuronidation. Increased incidence of rash-related withdrawal was seen when initial doses were higher and titration more rapid than recommended under Dosage and Administration.

Patients with a history of allergy or rash to other antiepileptic drugs: Caution is also required when treating patients with a history of allergy or rash to other antiepileptic drugs, as it was found in studies on the frequency of rash after treatment with lamotrigine that the rate of rash was approximately three to four times higher in patients with such a history than those without.

Restarting Apo-Lamotrigine Therapy: It is recommended that Apo-Lamotrigine not be restarted in patients who discontinued due to rash associated with prior treatment with Apo-Lamotrigine unless the potential benefits clearly outweigh the risks. If the decision is made to restart a patient who has discontinued Apo-Lamotrigine for any reason, the need to restart with the initial dosing recommendations should be assessed. The greater the interval of time since the previous dose, the greater consideration should be given to restarting with the initial dosing recommendations. If a patient has discontinued Apo-Lamotrigine for a period of more than 5 half-lives, it is recommended that initial dosing recommendations and guidelines be followed. The half-life of lamotrigine is affected by other concomitant medications.

Pregnant women: Apo-Lamotrigine should only be used during pregnancy if the expected benefits outweigh the potential risks. If therapy with Apo-Lamotrigine is considered necessary during pregnancy, the lowest possible therapeutic dose is recommended. Physiological changes during pregnancy may affect lamotrigine levels and/or therapeutic effect. There have been reports of decreased lamotrigine levels during pregnancy. Appropriate clinical management of pregnant women during lamotrigine therapy should be ensured.

Labour and Delivery: The effect of lamotrigine on labour and delivery in humans is unknown.

Breastfeeding: Lamotrigine has been reported to pass into breast milk resulting in total lamotrigine levels in infant's plasma, including neonates as well as older infants, of up to approximately 50% of the mother's plasma. Therefore, in some breast-fed infants, plasma concentrations of lamotrigine may reach levels at which pharmacological effects do occur. Because of the potential for adverse reactions from lamotrigine in nursing infants, breast-feeding while taking this medication is not recommended. Should a woman breast-feed while on therapy with lamotrigine, the infant should be monitored for adverse effects, such as sedation, rash and poor weight gain. Events including apnea, drowsiness, and poor sucking have been reported in infants who have been human milk-fed by mothers using lamotrigine; whether or not these events were caused by lamotrigine is unknown.

Geriatrics: As the pharmacokinetics in this age group do not differ significantly from a non-elderly adult population, no dosage adjustment from the recommended adult schedule is required.

DRUG INTERACTIONS

Uridine 5'-diphospho (UDP)-glucuronyl transferases (UGTs) have been identified as the enzymes responsible for metabolism of lamotrigine. Drugs that induce or inhibit glucuronidation may, therefore, affect the apparent clearance of lamotrigine. Strong or moderate inducers of the cytochrome P450 3A4 (CYP3A4) enzyme, which are also known to induce UGTs, may also enhance the metabolism of lamotrigine.

The effects of lamotrigine on the induction of specific families of mixed-function oxidase isozymes have not been systematically evaluated. Lamotrigine may induce its own metabolism, but the effect is modest and unlikely to have significant clinical consequences.

A number of drugs (see Drug-Drug Interactions) have been shown to have little or no effect on the concentration of lamotrigine. Coadministration of such drugs would generally not be expected to result in any clinical impact. However, consideration should be given to patients whose epilepsy is especially sensitive to fluctuations in concentrations of lamotrigine.

Drug-Behavioural Interactions:

Dependence Liability: No evidence of abuse potential has been associated with lamotrigine, nor is there evidence of psychological or physical dependence in humans.

Drug-Drug Interactions:

The drugs listed in this table are based on either drug interaction case reports or studies, or potential interactions due to the expected magnitude and seriousness of the interaction (i.e., those identified as contraindicated).

Antiepileptic Drugs (AEDs): The net effects of co-administration of lamotrigine with other antiepileptic drugs are summarized in table below. Additional details of these drug interaction studies are provided below.

AED	AED Plasma Concentration with Adjunctive LAM ¹	Lamotrigine Plasma Concentration with Adjunctive AEDs ²
Carbamazepine (CBZ) CBZ epoxide³	No significant effect Conflicting data	↓ 40%
Felbamate	Not evaluated	No significant effect
Gabapentin	Not evaluated	No significant effect
Lacosamide	Not evaluated	No significant effect
Levetiracetam	No significant effect	No significant effect
Oxcarbazepine	No significant effect	No significant effect
10-monohydroxy oxcarbazepine metabolite	No significant effect	Not evaluated
Perampanel	No significant effect ⁴	No significant effect
Phenobarbital	No significant effect	↓ 40%
Phenytoin (PHT)	No significant effect	↓ 40%
Pregabalin	No significant effect	No significant effect
Primidone	No significant effect	↓ 40%
Topiramate	No significant effect ⁴	No significant effect
Valproic Acid (VPA)	Decreased ⁵	↑ 200%
VPA + PHT and/or CBZ	Not evaluated	No significant effect
Zonisamide	Not evaluated	No significant effect

LAM = Lamotrigine.

¹ From adjunctive clinical trials and volunteer studies.

² Net effects were estimated by comparing the mean clearance values obtained in adjunctive clinical trials and volunteer studies.

³ Not administered, but an active metabolite of carbamazepine.

⁴ Slight increase, not expected to be clinically relevant.

⁵ Refer to Valproic Acid paragraph below for additional information.

↓ Decreased (induces lamotrigine glucuronidation)

↑ Increased (inhibits lamotrigine glucuronidation)

Lamotrigine does not affect the plasma concentrations of concomitantly administered AEDs that induce lamotrigine glucuronidation. Antiepileptic drugs (such as phenytoin, carbamazepine, phenobarbital, and primidone) that induce cytochrome P450 enzymes also induce UGTs and therefore, enhance the metabolism of lamotrigine by increasing the plasma clearance and reducing the elimination half-life of lamotrigine.

Carbamazepine: There have been reports of central nervous system events including dizziness, ataxia, diplopia, blurred vision, and nausea in patients taking carbamazepine following the introduction of lamotrigine tablet. These events usually resolve when the dose of carbamazepine is reduced.

Gabapentin: Based on a retrospective analysis of plasma levels of epileptic patients who received lamotrigine both with and without gabapentin, gabapentin did not appear to change the apparent clearance of lamotrigine. No information is available about the effects of gabapentin on C_{max} or AUC of lamotrigine.

Lacosamide: Plasma concentrations of lamotrigine were not affected by concomitant lacosamide in placebo-controlled clinical trials in patients with partial-onset seizures.

Levetiracetam: Potential drug interactions between levetiracetam and lamotrigine were assessed by evaluating serum concentrations of both agents during placebo-controlled clinical trials. In refractory epileptic patients receiving stable doses of lamotrigine, concomitant levetiracetam did not influence mean steady serum concentrations of lamotrigine. Similarly, lamotrigine did not influence the pharmacokinetics of levetiracetam.

Oxcarbazepine: The AUC and C_{max} of oxcarbazepine and its active 10-monohydroxy oxcarbazepine metabolite were not significantly different following the addition of oxcarbazepine to lamotrigine in healthy male volunteers compared to healthy male volunteers receiving oxcarbazepine alone. Limited clinical data suggest a higher incidence of headache, dizziness, nausea, and somnolence with co-administration of lamotrigine and oxcarbazepine compared to lamotrigine alone or oxcarbazepine alone.

Perampanel: In a population pharmacokinetic analysis of data from placebo-controlled clinical trials investigating adjunctive perampanel in patients with partial-onset and primary generalised tonic-clonic seizures, the highest perampanel dose evaluated increased lamotrigine clearance but is not considered to be clinically relevant.

Pregabalin: In patients with partial-onset seizures receiving combination lamotrigine and pregabalin 600 mg/day therapy, steady-state trough plasma concentrations of lamotrigine were not affected by pregabalin. The pharmacokinetics of pregabalin when co-administered with lamotrigine in this study appeared similar to historical values for pregabalin in healthy volunteers.

Topiramate: In studies in patients with epilepsy, topiramate resulted in no change in plasma concentrations of lamotrigine. Administration of lamotrigine resulted in an increase in plasma concentrations of topiramate.

Valproic Acid: Valproic acid reduces the plasma clearance and prolongs the elimination half-life of lamotrigine. However, the addition of lamotrigine did not affect the plasma concentration of valproic acid in patients receiving AEDs that induce lamotrigine glucuronidation in combination with valproic acid.

Zonisamide: In a study of patients with epilepsy, coadministration of zonisamide with lamotrigine had no significant effect on the pharmacokinetics of lamotrigine. Although the effects of lamotrigine on pharmacokinetics of zonisamide have not been studied, post-marketing cases have reported zonisamide toxicity when lamotrigine was administered. In both cases, positive dechallenge and rechallenge were observed.

Acetaminophen (Paracetamol): In a study in healthy volunteers, paracetamol 1 g reduced the plasma AUC and $C_{ss,min}$ of lamotrigine. The formation clearance of lamotrigine glucuronides was increased. Rarely, cases of seizures / drug inefficiency have been reported.

Aripiprazole: In a study of adult patients with bipolar I disorder, receiving an established regimen of lamotrigine, doses of aripiprazole were gradually increased and maintained for a short period. A reduction of $C_{\max}$ and AUC of lamotrigine was observed but not expected to be of clinical consequence.

Atazanavir/ritonavir: To initiate lamotrigine in patients already on the combination therapy of atazanavir/ritonavir, dose titration should follow the recommended guidelines based on whether lamotrigine is added to valproate (an inhibitor of lamotrigine glucuronidation), or to an inducer of lamotrigine glucuronidation, or whether lamotrigine is added in the absence of valproate or an inducer of lamotrigine glucuronidation.

To initiate atazanavir/ritonavir in patients already on maintenance doses of lamotrigine, the lamotrigine dose may need to be increased if another glucuronidation inducer is taken, but its dose may remain unchanged in the absence of another inducer. The lamotrigine dose needs to be decreased if atazanavir/ritonavir are discontinued.

Lopinavir/ritonavir: Caution is advised for the co-administration of lopinavir/ritonavir with lamotrigine. In patients receiving concomitant therapy with lopinavir/ritonavir, the treatment regimen recommended for concomitant medications that induce lamotrigine glucuronidation (without VPA) should be used.

Olanzapine: The AUC and $C_{\max}$ of lamotrigine was reduced following the addition of olanzapine to lamotrigine in healthy male volunteers compared to healthy male volunteers receiving lamotrigine alone, but this reduction in lamotrigine plasma concentrations is not expected to be clinically relevant.

Organic Cationic Transporter 2 (OCT 2): Lamotrigine is an inhibitor of renal tubular secretion via OCT 2 proteins. This may result in increased plasma levels of certain drugs that are substantially excreted via this route. Co-administration of lamotrigine with OCT 2 substrates (i.e. procainamide and metformin) with a narrow therapeutic index is not recommended.

Effect of Oral Contraceptives on Lamotrigine: Gradual transient increases in lamotrigine levels will occur during the week of no active hormone preparation (pill-free week) for women not also taking a drug that increases the clearance of lamotrigine (carbamazepine, phenytoin, phenobarbital, primidone, or rifampin). The increase in lamotrigine levels will be greater if the dose of lamotrigine is increased in the few days before or during the pill-free week. There have been post marketing reports of breakthrough seizures in women taking lamotrigine and hormonal contraceptives. Dosage adjustments will be necessary in most cases for women receiving oral contraceptive preparations.

Effect of Lamotrigine on Oral Contraceptives: The clinical significance of the observed hormonal changes on ovulatory activity is unknown. However, the possibility of decreased contraceptive efficacy in some patients cannot be excluded. Therefore, patients should be instructed to promptly report changes in their menstrual pattern (e.g., breakthrough bleeding).

Interactions with Other Hormonal Contraceptives or Hormone Replacement Therapy: The effect of other hormonal contraceptive preparations or estrogen-containing therapies, including hormone replacement therapy (HRT) on the pharmacokinetics of lamotrigine has not been evaluated, although the effect may be similar to oral contraceptive preparations. There have been reports of a possible interaction with estrogen HRTs which could lead to decreased efficacy of lamotrigine. Therefore, close clinical monitoring and similar dosage adjustments as for oral contraceptives may be necessary.

Rifampin: For patients taking rifampin with lamotrigine, follow the titration schedule for concomitant medications that induce lamotrigine glucuronidation (without VPA).

Risperidone: Following the co-administration of risperidone with lamotrigine, some reported somnolence compared to fewer when risperidone or lamotrigine was given alone.

Drug-Herb Interactions

Interactions with herbal products have not been established.

Drug-Laboratory Test Interactions

Lamotrigine has been reported to interfere with the assay used in some rapid urine drug screens, which can result in false positive readings, particularly for phencyclidine (PCP). A more specific alternative chemical method should be used to confirm a positive result.

ADVERSE REACTIONS

Serious skin rashes, including Erythema multiforme, Stevens Johnson Syndrome, and Toxic Epidermal Necrolysis (Lyell Syndrome) have been reported. Although the majority recover following drug withdrawal, some patients experience irreversible scarring and there have been rare cases of associated death.

Adverse experiences in patients receiving lamotrigine were generally mild, occurred within the first two weeks of therapy, and resolved without discontinuation of the drug.

Clinical Trial Adverse Reactions

Clinical trials are conducted under very specific conditions. The adverse reaction rates observed in the clinical trials; therefore, may not reflect the rates observed in practice and should not be compared to the rates in the clinical trials of another drug. Adverse reaction information from clinical trials may be useful in identifying and approximating rates of adverse drug reactions in real-world use.

Commonly Observed

The most commonly observed adverse experiences associated with the use of adjunctive therapy with lamotrigine (incidence of at least 10%) were dizziness, headache, diplopia, somnolence, ataxia, nausea, and asthenia.

Dizziness, diplopia, ataxia and blurred vision were dose-related and occurred more commonly in patients receiving carbamazepine in combination with lamotrigine than in patients receiving other AEDs that induce lamotrigine glucuronidation in combination with lamotrigine. Reduction of the daily dose and/or alteration of the timing of doses of concomitant antiepileptic drugs and/or lamotrigine may reduce or eliminate these symptoms. Clinical data suggest a higher incidence of rash in patients who are receiving concomitant valproic acid, or non-inducing AEDs.

Adverse Events Associated with Discontinuation of Treatment

Across all adult add-on studies, the most common adverse experiences associated with discontinuation of lamotrigine were rash, dizziness, headache, ataxia, nausea, diplopia, somnolence, seizure exacerbation, asthenia, and blurred vision.

Serious Adverse Events Associated with Discontinuation of Treatment

Rash accounted for almost half of the discontinuations due to serious adverse experiences. More rapid initial titration of lamotrigine, and concomitant use of valproic acid were associated with higher incidences of rash-related withdrawal in clinical studies.

Less Common Clinical Trial Adverse Reactions:

Less common clinical trial adverse reactions that occurred with a frequency $\geq 1\%$ and $< 2\%$ are listed below.

Body as a Whole: chest pain, chills

Cardiovascular: cerebral infarction, hot flashes, palpitations

Gastrointestinal: dry mouth

Musculoskeletal: joint disorder, myasthenia

Nervous System: dream abnormality, dysarthria, grand mal convulsion, hostility, hypertension, hypesthesia, mind racing, muscle spasm, sleep disorder

Metabolic: hyperkalemia, peripheral edema

Skin: acne, alopecia, erythema multiforme

Special Senses: eye disorder

Urogenital: amenorrhea, dysuria

Post-Market Adverse Reactions: In addition to the adverse experiences reported during clinical testing of lamotrigine, the following adverse experiences have been reported in patients receiving marketed lamotrigine and from worldwide investigational use. These adverse experiences have not been listed above, and data are insufficient to establish causation.

Body as a Whole: hypersensitivity reaction/hypersensitivity syndrome*, multi-organ failure, progressive immunosuppression, tiredness

* Rash has also been reported as part of Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS); also known as hypersensitivity syndrome. This condition is associated with a variable pattern of systemic symptoms.

Gastrointestinal: Esophagitis

Cardiac disorders: stress cardiomyopathy, atrioventricular block, ventricular fibrillation, ventricular tachycardia, QRS prolongation, ventricular extrasystoles, PR-prolongation

Hematologic and Lymphatic: Hematological abnormalities (anemia, agranulocytosis, aplastic anemia, disseminated intravascular coagulation, hemolytic anemia, leucopenia, neutropenia, pancytopenia, red cell aplasia thrombocytopenia), lymphadenopathy*, haemophagocytic lymphohistiocytosis (HLH) (very rare, see section Special Warnings and Precautions), pseudolymphoma**

*There have been reported cases of lymphadenopathy in the absence of hypersensitivity reactions, in patients with or without previous history, and taking lamotrigine alone or in combination with other medications (including other AEDs). Discontinuation of lamotrigine or, in some cases, dose reduction resolved the reaction.

**There have been reports of pseudolymphoma, including cutaneous pseudolymphoma, with gradual development or delayed onset within months of starting treatment with lamotrigine; in most reported cases, discontinuation of lamotrigine resolved the reaction.

Hepatobiliary Tract and Pancreas: elevations of liver function tests, hepatic dysfunction, including hepatic failure*, pancreatitis

*Hepatic dysfunction usually occurs in association with hypersensitivity reactions, but isolated cases have been reported without overt signs of hypersensitivity.

Immunologic: hypogammaglobulinaemia*, lupus-like reaction, vasculitis

*There have been rare reports of reduced serum immunoglobulin levels in patients receiving lamotrigine. In some cases, discontinuation of lamotrigine with or without administration of intravenous immunoglobulin resolved the reaction. Serum immunoglobulin levels should be measured in lamotrigine-treated patients who experience recurrent infections.

Lower Respiratory: apnea

Metabolism and nutrition disorders: Hyponatremia

Musculoskeletal: rhabdomyolysis (observed in patients experiencing hypersensitivity reactions)

Neurologic: aggression, aseptic meningitis*, exacerbation of parkinsonian symptoms**, extrapyramidal effects, choreoathetosis, hallucinations, movement disorders (such as tics and unsteadiness), nightmares

*There have been rare reports of aseptic meningitis in patients taking lamotrigine alone or in combination with other AEDs. In some cases, discontinuation of lamotrigine resolved this reaction.

**There have been reports that lamotrigine may worsen parkinsonian symptoms in patients with pre-existing Parkinson's disease, and isolated reports of extrapyramidal effects and choreoathetosis in patients without this underlying condition.

Oral Contraceptives: There have been post marketing reports of breakthrough seizures in patients taking lamotrigine and hormonal contraceptives. In some cases, reduced serum lamotrigine levels during co-administration with a hormonal contraceptive have been documented. Most cases have responded to lamotrigine dose increase.

Renal and Urinary disorders: Tubulointerstitial nephritis* (may occur in association with uveitis)

*Tubulointerstitial nephritis usually occurs in association with hypersensitivity reactions, but isolated cases have been reported without overt signs of hypersensitivity. In the absence of hypersensitivity, discontinuation of lamotrigine and treatment with steroids resolved the reaction in some cases.

Skin: Photosensitivity reaction

OVERDOSAGE AND TREATMENT

Acute overdoses involving quantities up to 15 g have been reported for lamotrigine, some of which have been fatal. In general, overdose has resulted in symptoms including nystagmus, ataxia, grand mal convulsions, impaired consciousness coma and intraventricular conduction delay (QRS broadening).

There are no specific antidotes for Apo-Lamotrigine. Following a suspected overdose, immediate hospitalization of the patient is advised. In the event of a very recent overdose/ingestion of a potentially life-threatening amount of the drug, emesis may be induced if indicated. General supportive care is indicated, including frequent monitoring of vital signs and close observation of the patient. It is uncertain whether hemodialysis is an effective means of removing lamotrigine from the blood.

STORAGE CONDITIONS:

Store below 30°C. Protect from light.

AVAILABILITY OF DOSAGE FORMS

Apo-Lamotrigine 25 mg: each white, shield-shaped, flat-faced, radial edged tablet, scored and engraved “25” on one side, other side “APO” contains 25mg of lamotrigine. Available in HDPE bottles of 30’s and 100’s. Apo-Lamotrigine 100 mg: each peach, shield-shaped, flat-faced, bevelled edged tablet, scored and engraved “LAM over 100” on one side, other side “APO” contains 100mg of lamotrigine. Available in HDPE bottles of 30’s and 100’s.

MANUFACTURER’S NAME and ADDRESS:

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DATE OF REVISION:

11 May 2026

CA0082C-4